

The limitations of estimating implied densities from option prices

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Bakshi *et al.* (Stock Return Characteristics, Skew Laws, and the Differential Pricing of Individual Equity Options. *Rev. Financ. Stud.*, 2003, **16**(1), 101–143) develop a novel method for estimating a stock's τ -period risk-neutral return moments from option prices. Their result relies on the integration of all OTM calls and puts to find the prices of a quadratic, cubic, and quartic contract; from which risk-neutral estimates of the 2nd, 3rd, and 4th moments are derived. Dennis and Mayhew (Risk-neutral skewness: Evidence from stock options. *J. Financ. Quant. Anal.*, 2002, **37**(3), 471–493) point out that error is introduced into the BKM estimation in practice since options do not span a continuum and the range of moneyness and distance between strikes varies across firms and maturities. We use Monte Carlo simulation to show that the standard approach of Dennis and Mayhew (Risk-neutral skewness: Evidence from stock options. *J. Financ. Quant. Anal.*, 2002, **37**(3), 471–493) used within Conrad *et al.* (Ex ante skewness and expected stock returns. *J. Finance.*, 2013, **68**(1), 85–124) and other leading empirical works leads to very biased, and in some cases inconsistent, estimates of BKM implied moments. More so, we prove the particularly concerning result that estimation error using the Dennis and Mayhew (Risk-neutral skewness: Evidence from stock options. *J. Financ. Quant. Anal.*, 2002, **37**(3), 471–493) methodology increases dramatically when a stock's actual return distribution is non-normal or the firm's stock price is not *exactly* at-the-money. Fortunately, the 2-step interpolation method developed by DeMiguel *et al.* (Improving portfolio selection using option-implied volatility and skewness. *J. Financ. Quant. Anal.*, 2013, **48**(6), 1813–1845) is largely robust to the estimation errors we document. Based on our simulation results, we recommend the use of the DeMiguel *et al.* (Improving portfolio selection using option-implied volatility and skewness. *J. Financ. Quant. Anal.*, 2013, **48**(6), 1813–1845) methodology to estimate the 3rd and 4th implied moments, and either the DeMiguel *et al.* (Improving portfolio selection using option-implied volatility and skewness. *J. Financ. Quant. Anal.*, 2013, **48**(6), 1813–1845) methodology or the methodology of Fukasawa *et al.* (Model-free implied volatility: From Surface to Index. *Int. J. Theor. Appl. Finance*, 2011, **14**(4), 433–463) to estimate the 2nd implied moment (implied volatility). In addition, we argue that the economically large and robust results of our MC simulations and brief empirical study indicate that the findings of prior studies which use the estimation methodology of Dennis and Mayhew (Risk-neutral skewness: Evidence from stock options. *J. Financ. Quant. Anal.*, 2002, **37**(3), 471–493), followed by Conrad *et al.* (Ex ante skewness and expected stock returns. *J. Finance.*, 2013, **68**(1), 85–124), should be reevaluated.

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JEL Classifications: C140, C150, G100

1. Introduction

Financial options allow market participants to observe information pertinent to the underlying firm that is unavailable without such contracts. A particular area of focus in the

options literature has been accurately extracting a stock's τ -period implied return density from observable option prices. This focus has led to the development of various popular approaches to estimate a stock's return density. If the process which governs a stock's return is known, then one may use calibration to find option-implied densities. However, if the return generating process were known, then there would be little intrigue in investigating a firm's implied density in the

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first place! Further, committing to a return model, such as a Geometric Brownian Motion (GBM) or a stochastic volatility process such as the Heston (1993) Model, requires the large assumption that the process is, in fact, correct; placing significant restriction on return dynamics. Thus, the calibration of parametric models has decreased in popularity in recent years. Instead, options researchers have increasingly focused on non-parametric or ‘model-free’ methods to estimate return moments.[†] Bakshi *et al.* (2003) (hereafter referred to as BKM) author the seminal paper on the non-parametric estimation of the implied moments of a stock’s underlying return distribution. They show that under the assumption of risk-neutrality, the second, third, and fourth implied moments of a stock’s returns can be estimated utilizing the full series of option contracts for a given tenor. The method involves integrating along a continuum of OTM puts and calls and finding the prices of a quadratic, cubic, and quartic τ -period payoff-contract.[‡] By imposing the assumption of risk-neutrality, the implied moments are estimates taken from the firm’s risk-neutral density function obtained from the firm’s option prices. While the risk-neutral density function is not the same as the *actual* density function, Dennis and Mayhew (2002) show that there is a one-to-one monotonic mapping between the risk-neutral density function and the slope of the implied volatility curve. Furthermore, Dennis and Mayhew (2002) show the importance of this, as positively sloped implied volatility curves, in which the implied volatility of out-of-the-money calls is greater than that of in-the-money calls, directly imply positive skewness in the stock’s risk-neutral density.[§] Thus, the risk-neutral τ -period moments implied by a firm’s τ -period option prices do provide information about a firm’s expected level of return dispersion, skewness, and kurtosis.

While BKM provides an elegant theory to estimate a firm’s option-implied return density, there are several obstacles to applying the BKM methodology in practice. The first two sources of bias mentioned are nicely discussed and investigated by Dennis and Mayhew (2002), while the others have yet to be quantified to the best of our knowledge. First, as Dennis and Mayhew (2002) note, a firm’s τ -period OTM option contracts do not span the continuum $[0, \infty)$ as required by BKM, but rather have discrete strikes with limited ranges of moneyness. Oftentimes, strikes are issued in \$1, \$5, or even \$10 intervals. Second, particularly when there is a small number of observed options, the number of options and range of moneyness observed for OTM calls versus OTM puts can lead to a bias in which the left or right-side of the distribution receives significantly more weight in determining the BKM implied moments. This can lead to large biases, particularly in implied skewness. We denote these first two sources of bias which stem from a finite number of strike prices observed in practice as the ‘*discretization bias*’.

The third source of bias, to the best of our knowledge, is uniquely recognized and estimated within our research. The

BKM methodology assumes the existence of an exactly at-the-money (ATM) strike price. While this is obvious when examining the BKM formulas, the magnitude of bias in implied moment estimates when the firm’s stock price is not *exactly* at-the-money is shocking. In practice, unlike a hand-crafted simulation, when examining the options chain of a security for any given tenor, it is incredibly unlikely to have a strike price which is *exactly* equal to the underlying stock price. We show that the BKM estimation methodology of Dennis and Mayhew (2002) relies upon and is extremely sensitive to this very unrealistic assumption of an *exactly* at-the-money strike. We appropriately denote this third source of bias the ‘*ATM bias*’.

Finally, the BKM method is, by its very design, non-parametric in nature; and thus the mechanics of the underlying return generating process itself should not bias the BKM implied moments. However, the biases noted above become exacerbated in the case that the underlying return generating process is highly non-normal. This is, in some sense, the most significant issue with the application of the Dennis and Mayhew (2002) estimation of the BKM methodology as the very firms for which we are most interested in getting accurate estimates of the implied moments are those which have the most non-normal returns; and thus are also those with the most biased BKM implied moment estimates. Thus, we examine the accuracy of BKM moment estimates under the case of normally distributed returns using the model of Black and Scholes (1973) and under the case of non-normally distributed returns using the Heston (1993) option pricing model in which we simulate returns for a typical firm with slightly left-skewed and leptokurtotic returns.

We first run a detailed simulation experiment to isolate the impact of each of the above biases. As mentioned above, we simulate options using either the Black and Scholes (1973) or Heston (1993) option pricing models and alter the number of contracts ‘observed’ and spot price of the underlying security. We follow two different, commonly used procedures to calculate BKM implied moments and examine how well they mitigate market frictions.

First, we test the approach introduced by Dennis and Mayhew (2002) and utilized by Conrad *et al.* (2013) in which only a symmetric range of observed options are used in the BKM calculation and a trapezoidal approximation is used to approximate the integration. We contrast this with the more modern and elegant approach utilized by DeMiguel *et al.* (2013), in which all observed OTM options are used and a cubic spline is fitted to the implied volatilities to create synthetic prices for a continuum of options prices. We demonstrate via simulation that the trapezoidal approximation methodology of Conrad *et al.* (2013) in which only a symmetric range of OTM puts and calls are used leads to extremely significant implied moment biases in the case of a firm without an *exactly* at-the-money strike price for the tenor observed or a firm whose returns are even slightly non-normally distributed (a typical firm in the cross-section). In contrast, these biases are greatly, if not altogether mitigated by the approach of DeMiguel *et al.* (2013).

In regards to the estimation of the 2nd return moment, we also test the popular model-free implied volatility estimation methodologies of Jiang and Tian (2005) as well as

[†] Dennis and Mayhew 2002 have a short but excellent discussion of this literature.

[‡] It also requires that the payoff function has bounded expectations, which is non-problematic as the payoffs have a minimum value of 0 and a maximum value of $\max(S_\tau - K, 0)$ in the case of a call and $\max(K - S_\tau, 0)$ in the case of a put.

[§] The similar, but opposite relationship holds for puts.

Fukasawa *et al.* (2011). We find that the methodology of Jiang and Tian (2005) provides more accurate, and certainly more consistent, estimates of implied volatility than the BKM methodology of Conrad *et al.* (2013), but does exhibit slight downward biases in a firm's implied volatility estimate when a firm has no exactly at-the-money strike price or non-normally distributed returns. However, we find that the model-free implied volatility estimation technique of Fukasawa *et al.* (2011) leads to extremely accurate implied volatility estimates, on par with those of the DeMiguel *et al.* (2013) approach to extract BKM implied volatility. Thus, for researchers and practitioners only focused on the estimate of implied volatility (rather than the second, third, and fourth return moments), we recommend either the use of the model-free implied volatility methodology of Fukasawa *et al.* (2011) or the DeMiguel *et al.* (2013) methodology based on personal preference, as both lead to extremely robust, accurate estimates of implied volatility – even for firms with no at-the-money option or those with non-normally distributed returns.

A final important implication of our simulation results is that the inferences of some leading empirical asset-pricing works, which rely upon the estimation of BKM implied moments using the Dennis and Mayhew (2002) approach, could be flawed. Specifically, the return inferences of Conrad *et al.* (2013) relate the cross-section of stock returns to firms' option-implied moments using the methodology of Dennis and Mayhew (2002) to extract BKM implied moments. The return inferences made by Conrad *et al.* (2013) may not be reliable; as the option-implied moments themselves used to predict cross-sectional returns are highly biased, and in some cases inconsistent, using the approach of Dennis and Mayhew (2002) followed by Conrad *et al.* (2013). More so, the 'omitted variables' driving these option-implied moment biases have to do with a firm's option microstructure, and *actual* return generating process, both of which are quite complicated and tough, it not altogether impossible, to disentangle for any given firm in the cross-section.

Due to the issues inherent in the Conrad *et al.* (2013) approach, we analyze how informative the implied moments are in explaining subsequent returns when using the approach of Conrad *et al.* (2013) and the approach of DeMiguel *et al.* (2013). Following the tests in Conrad *et al.* (2013), we create tercile portfolios based on the ranking of each of the implied moments and examine subsequent monthly raw and characteristic adjusted returns which control for size and book-to-market effects. We also examine the informational content of the higher moments once volatility is controlled for using double-sorted portfolios. Our main results of these tests highlight concerns regarding the inferences made by Conrad *et al.* (2013). Put simply, we demonstrate that inferences regarding implied moment portfolio returns differ somewhat based on the estimation methodology employed. Since the approach of DeMiguel *et al.* (2013) clearly leads to much lower biases in estimating BKM implied moments in our robust simulation tests, we recommend that our results using the estimation approach of DeMiguel *et al.* (2013) be relied upon in favor of our results using the methodology of Dennis and Mayhew (2002) followed by Conrad *et al.* (2013). In

addition, we recommend that other studies which utilize the approach of Conrad *et al.* (2013), and have important implications for empirical asset pricing, be replicated for accuracy using the approach of DeMiguel *et al.* (2013).

Of course, since the magnitude of the estimation error in the Dennis and Mayhew (2002) estimation approach followed by Conrad *et al.* (2013) was not previously realized; it is more than forgivable that a stream of papers have implemented these sensible, but very biased techniques. We therefore want to clearly note that we do not wish to critique the choice of methodology of any researchers involved, as the approach of Conrad *et al.* (2013) was sensible in and all regards given the information available at the time. However, we strongly recommend the approach of DeMiguel *et al.* (2013) going forward, as it eliminates or nearly eliminates the significant biases we document within the approach of Conrad *et al.* (2013) and Dennis and Mayhew (2002).

Our results contribute to the broad options literature as well as the empirical asset pricing literature. As BKM is the seminal paper in the non-parametric estimation of implied moments, our paper relates to a stream of options literature that has utilized the BKM methodology without regard to biases introduced in the estimation procedure. The biases we identify, along with those already identified in the literature by Dennis and Mayhew (2002), show that estimated BKM implied moments are highly influenced by the firm's options microstructure when using simplistic estimation procedures. Based on this, care needs to be taken to ensure that a robust procedure is used in estimating BKM implied moments. While we raise significant concerns regarding inferences made based on the more simplistic approach of Conrad *et al.* (2013), our results also indicate that the slightly more complex 2-step interpolation approach of DeMiguel *et al.* (2013) is a very robust approach in practice.

Our results are important to a modern, empirical asset pricing literature that is increasingly fascinated by the relation between option-implied moments and stock returns. We show that the main result in this literature of Conrad *et al.* (2013) is greatly dependent on firms' option microstructures and the accompanying biases we document. Furthermore, when a more robust and accurate estimation procedure to extract BKM implied densities is utilized, the limited results of Conrad *et al.* (2013) which we replicate are somewhat altered. The implication is that the leading result in the literature which relates option-implied densities to the cross-section of stock returns is heavily dependent on a flawed estimation technique. This is a significant finding for the empirical asset pricing literature, as inferences relating implied moments to the cross-section of stock returns should carefully be readdressed using the methodology of DeMiguel *et al.* (2013).

The rest of the paper is structured as follows. We first outline the different approaches to calculate the BKM moments and outline our simulation methodology in section 2. We detail the data used in the empirical portion of our study in section 3. We discuss the results of our simulation study and the analysis of each approach related to the cross-section of returns in section 4 and conclude in section 5.

2. Simulation methodology

In order to show the magnitude of the ‘discretization bias’ and the ‘ATM bias’ for firms with varying underlying return generating processes, we simulate a ‘typical’ firm with normally distributed returns, and a ‘typical’ firm with non-normally distributed returns. Following the intuition set forth in Dennis and Mayhew (2002) we first simulate Black–Scholes European options prices for the normal case, as we know the prices in Black–Scholes are log-normally distributed and thus log-returns are normally distributed. As such, the actual skewness of the log-returns over any period τ must be 0, the kurtosis 3, and the volatility equal to $\sigma\sqrt{\tau}$ (where σ is annualized volatility). By changing the distance between a hypothetical firm’s strike prices and the range of moneyness in the firm’s contracts, we are able to accurately show the magnitude of the ‘discretization bias’. To show the impact of the ‘ATM bias’ we simply alter the spot price of the underlying stock between the two nearest surrounding strike prices.

Now, the above simulations with normally distributed returns are perhaps the least interesting scenarios as the primary purpose of a non-parametric technique such as BKM is to estimate the implied moments for firms who *likely* have highly non-normal return densities. To test the estimation error for such firms, we resort to Monte Carlo simulation and a return generating process which allows for skewness and leptokurtosis. The return generating process we use is a discretization of the Heston (1993) Stochastic Volatility model. In Heston’s model the change in variance at period t , dV_t , and change in the stock price at period t , dS_t follow the process[†]:

$$dS_t = rS_t dt + \sqrt{V_t}S_t \left[\rho dW_t^{(1)} + \sqrt{1 - \rho^2} dW_t^{(2)} \right] \quad (1)$$

$$dV_t = \kappa(\theta - V_t) dt + \sigma_v \sqrt{V_t} dW_t^{(1)} \quad (2)$$

Equation (1) shows the change in the firm’s stock price, at any period t where r is the risk-free rate, ρ the instantaneous correlation between the variance process and the stock price process, and $dW_t^{(1)}$ and $dW_t^{(2)}$ are independent Brownian motion processes. In equation (2), κ is the mean-reversion parameter, θ the long-run variance, and σ_v is a parameter governing the volatility of the variance process.[‡]

Now, in testing the estimation error of firms with non-normal return processes we use a discretization of Heston’s Stochastic volatility, thus equation (1) becomes:

$$S_t - S_{t-1} = rS_{t-1}\Delta t + \sqrt{V_{t-1}}S_{t-1} \times \left[\rho \Delta W_t^{(1)} + \sqrt{1 - \rho^2} \Delta W_t^{(2)} \right], \quad (3)$$

And, similarly equation (2) becomes:

$$V_t - V_{t-1} = \kappa(\theta - V_{t-1})\Delta t + \sigma_v \sqrt{V_{t-1}} \Delta W_t^{(1)}. \quad (4)$$

[†] Broadie and Kaya (2006) have an excellent discussion of Heston’s SV model. Broadie and Kaya (2006) also discuss the reasons for and magnitude of the Euler discretization bias inherent in the discrete time model. Finally, we follow the notation of Broadie and Kaya (2006) in describing Heston’s stochastic volatility model in equations (1–4).

[‡] The volatility of the variance is commonly called the ‘vol. of vol.’ parameter.

Broadie and Kaya (2006) show that the Euler discretization of Heston’s SV process biases expected option prices. Not surprisingly, the bias decreases with the number of time steps and increases with the time-to-maturity, τ (or equivalently, it decreases as Δt decreases). For example, Broadie and Kaya (2006) show that with a 3200 time-step discretization, or Δt of 1/3200, and with reasonable values of κ , θ , ρ , r , and σ_v , an ATM call option with a time-to-maturity, τ , of 5 years and a current stock price of \$100 has a bias of \$.0161 with a true option price of \$6.8061. Increasing Δt to 1/100 dramatically increases the bias in the call option’s price to \$.1543. To combat this problem, we do two things. First, we of course use a large number of time steps (5000) in our simulations. Second, we examine a hypothetical contract with a 1-month tenor. By increasing the number of time steps and having a much shorter tenor on the option contract, we have a sufficiently small Δt to minimize any impact from the Euler discretization bias inherent in this discrete-time approach. By examining a hypothetical contract with a 1-month tenor, we also ensure that we are examining options that are likely of the most interest to traders and academics alike, the front-month options contract. Finally, Broadie and Kaya (2006) note that the Euler Discretization error in V_t is significantly less troubling when $\frac{4\theta\kappa}{\sigma_v} > 1$ (as for large values of this fraction the Euler discretization of V_t given an earlier draw from the variance process, V_u is approximately distributed χ^2 with $\frac{4\theta\kappa}{\sigma_v} - 1$ degrees of freedom plus a trivial non-centrality parameter, so long as $t - u$ is sufficiently small and/or κ is sufficiently small). Thus, we purposefully pick parameters of θ , κ , and σ_v to make the ratio $\frac{4\theta\kappa}{\sigma_v}$ greater than 1. Each of these factors is designed to minimize the impact of any Euler discretization bias in our simulated prices so that we are able to get precise measurements of a firm’s European options contracts and the ensuing BKM implied moments.

In testing the Heston (1993) model, we are able to use the closed-form solution to simulate the call prices; however to get a measure for the underlying distribution to benchmark against, we need to resort to Monte Carlo simulation. In performing our simulation, we simulate 1 million paths, each with 5000 time steps for the underlying stock price with the underlying stochastic volatility process following equations (3) and (4). For each path, we are able to calculate return over the 1-month period and use the 1 million simulated paths to create the empirical return distribution. This empirical distribution is then used to calculate the hypothetical firms’ actual volatility, skewness and kurtosis. This allows us to accurately compare how closely the implied moments of the two estimations of the BKM methodology reflect the actual sample moments given differences in strike price intervals and moneyness ranges for a hypothetical firm with non-normal returns.

Below, we briefly outline the BKM model[§]. Assuming a continuum of strike prices the price of a quadratic contract can be calculated from the following equation:

$$V(t, \tau) = \int_{S(t)}^{\infty} \frac{2(1 - \ln[\frac{K}{S(t)})]}{K^2} C(t, \tau; K) dK$$

[§] See Bakshi et al. (2003) for full details.

$$+ \int_0^{S(t)} \frac{2(1 + \ln[\frac{S(t)}{K}])}{K^2} P(t, \tau; K) dK \quad (5)$$

which is the price of a τ -period volatility contract at time t . K represents the strike price of each option and $S(t)$ the stock price at time t . $C(t, \tau; K)$ is the price of a τ -period call option with underlying strike price K at the current time t . Similarly, we can also calculate the price of a τ -period cubic contract, W , and quartic contract, X

$$W(t, \tau) = \int_{S(t)}^{\infty} \frac{6(\ln[\frac{K}{S(t)}]) - 3(\ln[\frac{K}{S(t)}])^2}{K^2} C(t, \tau; K) dK \\ - \int_0^{S(t)} \frac{6(\ln[\frac{S(t)}{K}]) + 3(\ln[\frac{S(t)}{K}])^2}{K^2} P(t, \tau; K) dK \quad (6)$$

$$X(t, \tau) = \int_{S(t)}^{\infty} \frac{12(\ln[\frac{K}{S(t)}])^2 - 4(\ln[\frac{K}{S(t)}])^3}{K^2} C(t, \tau; K) dK \\ + \int_0^{S(t)} \frac{12(\ln[\frac{S(t)}{K}])^2 + 4(\ln[\frac{S(t)}{K}])^3}{K^2} P(t, \tau; K) dK \quad (7)$$

After obtaining the prices of a quadratic, cubic, and quartic contract, one can solve for the parameter $\mu(t, \tau)$:

$$\mu(t, \tau) = e^{rt} - 1 - \frac{e^{rt}V(t, \tau)}{2} - \frac{e^{rt}W(t, \tau)}{6} - \frac{e^{rt}X(t, \tau)}{24}. \quad (8)$$

and finally, calculate the risk-neutral implied volatility, skewness and kurtosis.

$$\sigma(t, \tau) = \sqrt{\frac{e^{rt}V(t, \tau) - \mu(t, \tau)^2}{\tau}} \quad (9)$$

$$skew(t, \tau) = \frac{e^{rt}W(t, \tau) - 3\mu(t, \tau)e^{rt}V(t, \tau) + 2\mu(t, \tau)^3}{[e^{rt}V(t, \tau) - \mu(t, \tau)^2]^{3/2}} \quad (10)$$

$$kurt(t, \tau) = \frac{e^{rt}X(t, \tau) - 4\mu(t, \tau)e^{rt}W(t, \tau) + 6e^{rt}\mu(t, \tau)^2V(t, \tau) - 3\mu(t, \tau)^4}{[e^{rt}V(t, \tau) - \mu(t, \tau)^2]^2} \quad (11)$$

In this study we analyze the BKM implied moment estimates using two different leading methodologies. The first methodology we use is the approach of Dennis and Mayhew (2002) used within Conrad *et al.* (2013) and the second the approach is that used within DeMiguel *et al.* (2013).

The Conrad *et al.* (2013) estimation procedure naturally suffers from the shortcomings of the estimation approach which it follows, that of Dennis and Mayhew (2002). Dennis and Mayhew (2002) examine the impacts of the discretization bias and determine that the magnitude of the bias is not substantial. Dennis and Mayhew (2002) do note that there is a substantial concern around the symmetry of put and call options. As such, they recommend using only a symmetric range of options and omitting any remaining strikes. Based on this, the Dennis and Mayhew (2002) approach requires that there must be an equal number of out-of-the-money puts and calls (or that one limits observed options strikes so that this is

the case) and requires that there are at least two out-of-the-money put and call strikes to allow estimates of the BKM integrals. This approach has two advantages: (1) it is simplistic to implement and (2) it only utilizes actual, observed option prices to estimate the BKM implied moments. The very large disadvantage of this approach is that it is not fully consistent with the intuition of the BKM formula as dropping far OTM observed option prices results in dropping information about the firm's underlying distribution, leading to particularly large biases in higher moment estimation. In addition, implied moment estimates of the approach suffer in all instances in which the firm's stock price is not *exactly* at-the-money.

In contrast, the approach used by DeMiguel *et al.* (2013) involves a two-step process. First using all observed OTM options, option prices are inverted to implied volatilities using the Black and Scholes (1973) option pricing model. A cubic spline is then fit to the resulting BS implied volatilities to give a full continuum of strike prices, where the implied volatilities outside the observed strike range are assumed to be equal to the implied volatility of the most extreme observation. The resulting BS implied volatilities are then converted back into option prices using the Black and Scholes (1973) option pricing model. Using the resulting option prices, DeMiguel *et al.* (2013) then directly calculate the equations outlined in BKM. This approach has the advantage of being fully consistent with the assumptions underlying the BKM model as there is a continuum of option prices synthetically created. This also completely resolves the ATM bias as there will now exist a (synthetically created) option with a strike that is *exactly* at-the-money. The obvious disadvantage to the approach is that by using statistical methods to infer a continuum of option prices; the vast majority of options used in the BKM integrations are synthetically created rather than actually observed.

At this point we should also note that some other researchers, even prior to DeMiguel *et al.* (2013), utilize sophisticated interpolation methods very similar in spirit to that of DeMiguel *et al.* (2013); although a great deal of applied research utilizing the BKM model tends to use very inaccurate interpolation methods similar to the approach of Dennis and Mayhew (2002). For example; Chang *et al.* (2012) form what appears to be a very accurate interpolation of the BKM approach as Chang *et al.* (2012) follow the interpolation procedure of Jiang and Tian (2005) to interpolate and form a continuum of IVs- and the Jiang and Tian (2005) interpolation procedure of IVs is extremely similar to the DeMiguel *et al.* (2013) procedure described in the paragraph above. Thus, some prior empirical research does an exceedingly good job of accurately extracting BKM implied moment estimates. However, our primary contribution is to compare the magnitude of bias, consistency, and tradeoffs between the two most commonly applied approaches.

Ultimately, we argue that the methodology which minimizes biases and maximizes consistency in the implied moments across a range of real-world scenarios including varying ranges of moneyness and distances between strikes; as well as normal and non-normal underlying return distributions are preferred.

We compare the BKM implied moments from the Dennis and Mayhew (2002) methodology followed by Conrad *et al.* (2013) and the DeMiguel *et al.* (2013) estimation methodology to the true moments of the underlying return generating process to determine the accuracy of each methodology in situations representative of those observed in the real world. In doing so, we are able to attest to which methodology offers a superior and more robust estimation procedure of the BKM implied moments.

In the case of the implied volatility estimate, we also compare the model-free implied volatility estimation methodologies of Jiang and Tian (2005) and Fukasawa *et al.* (2011) alongside the BKM implied volatility estimates following the approach of Conrad *et al.* (2013) and that of DeMiguel *et al.* (2013). Since many options traders, portfolio managers, and researchers are most concerned with the implied volatility estimate, and thus many do not use the BKM approach, we felt it pertinent to compare popular model-free implied volatility estimation methodologies alongside the two BKM approaches. However, due to the fact that the performance of model-free implied volatility models is not the focus of our research, and the fact that both the Jiang and Tian (2005) and Fukasawa *et al.* (2011) algorithms to extract implied volatility are sufficiently complex, we omit a step-by-step description of each herein, but provide a reference for interested readers below.[†]

3. Data

Our option price dataset is from Historical Option Prices. We randomly checked the dataset for accuracy against other leading options databases such as Optionmetrics and found the data to be high quality and accurate. We begin with daily option price data for all out-of-the-money calls and out-of-the-money puts from February 2002 to April 2016. Following the standard methodology, closing prices are constructed as midpoint averages of the daily last bid and ask price and we calculate implied volatilities (IVs) using the Black and Scholes (1973) model. We add filters to ensure that the close is greater than \$0.50, there is non-zero open interest in the option contract, an IV exists and is less than 1 (100% implied volatility), and that the option has a moneyness between 0.5 and 1.5. Furthermore, we require that for each given option-maturity series there are at least two puts and two calls that survive the previous filters, as the BKM methodology requires such.

Data on stock returns are obtained from the Center for Research in Security Prices (CRSP). We obtain CRSP

monthly stock returns from 2002 through 2016 for all individual securities with common shares outstanding. In addition, we obtain balance sheet data from COMPUSTAT to calculate book-to-market ratios following the common methodology of Fama and French (1993).

4. Results

In this section, we detail the results of our simulation study, covering results of tests of the discretization bias and ATM bias for both the normal (Black and Scholes 1973) case and the non-normal (Heston 1993) case. We then follow this with our brief empirical study which replicates the primary tests within Conrad *et al.* (2013) and shows that the Dennis and Mayhew (2002) and DeMiguel *et al.* (2013) approaches to estimate BKM implied moments lead to different inferences regarding the impact of firms' implied moments on the cross-section of stock returns.

4.1. Simulation results

To gain a primary intuition of the limitations in BKM estimation, we first simulate option prices using the most simplified and well-known model to determine actual firm returns; the Black and Scholes (1973) model. We create Black–Scholes European put and call prices for a firm with a stock price of \$50, annualized volatility of 40%, time-to-maturity of 1-month (1/12 year), and set the risk-free rate at 2%. For this parameterization, we calculate put and call prices for all strike prices between \$10 and \$90 at 1-cent intervals. We take the 8001 resulting put and call prices which span a wide moneyness of .2 to 1.8 to be a close enough representation of the continuous series of strikes required by the BKM model. Therefore, we expect that the calculated moments using these 8,001 strikes at 1-cent intervals should converge to the true underlying distribution moments. Since the log-return of the Black–Scholes process is normally distributed, the true values of the underlying moments are 40% for annualized volatility, 0 for skewness, and 3 for kurtosis. Of course, no firm would have options listed at 1-cent intervals as there simply would not typically be demand for contracts at an obscure price such as \$33.77. Nonetheless, this is a nice theoretical baseline scenario in which BKM estimation using the methodology of Dennis and Mayhew (2002) applied by Conrad *et al.* (2013) can be shown to be highly accurate. As we move to more realistic scenarios, the biases of the Dennis and Mayhew (2002) estimation methodology to extract BKM implied moments become incredibly apparent; both statistically and economically.

For all of our simulation tests, the results using the methodology of Dennis and Mayhew (2002) followed by Conrad *et al.* (2013) to extract BKM implied moments are abbreviated as 'BKM', the DeMiguel *et al.* (2013) results to extract BKM implied moments are abbreviated as 'DPUV', the Fukasawa *et al.* (2011) model-free implied volatility estimate is reported as 'Fukasawa', and the Jiang and Tian (2005) model-free implied volatility estimate is reported as 'Jiang' within the chart keys listed in figures 1–4.

[†] Jiang and Tian (2005) have a very clear explanation of their model-free implied volatility estimation methodology with formulas included and steps clearly described. Please see Jiang and Tian (2005), pgs. 1310–1316. In addition, Fukasawa *et al.* (2011) have a wonderful description of their model-free implied volatility estimation methodology laid out in a step-by-step approach (steps 1–4) within their paper, which includes an illustrative example of the exact calculations and steps necessary to calculate the estimate using a sample of Nikkei 225 index options. Please see Fukasawa *et al.* (2011), pgs. 433–445 for this very informative step-by-step explanation.

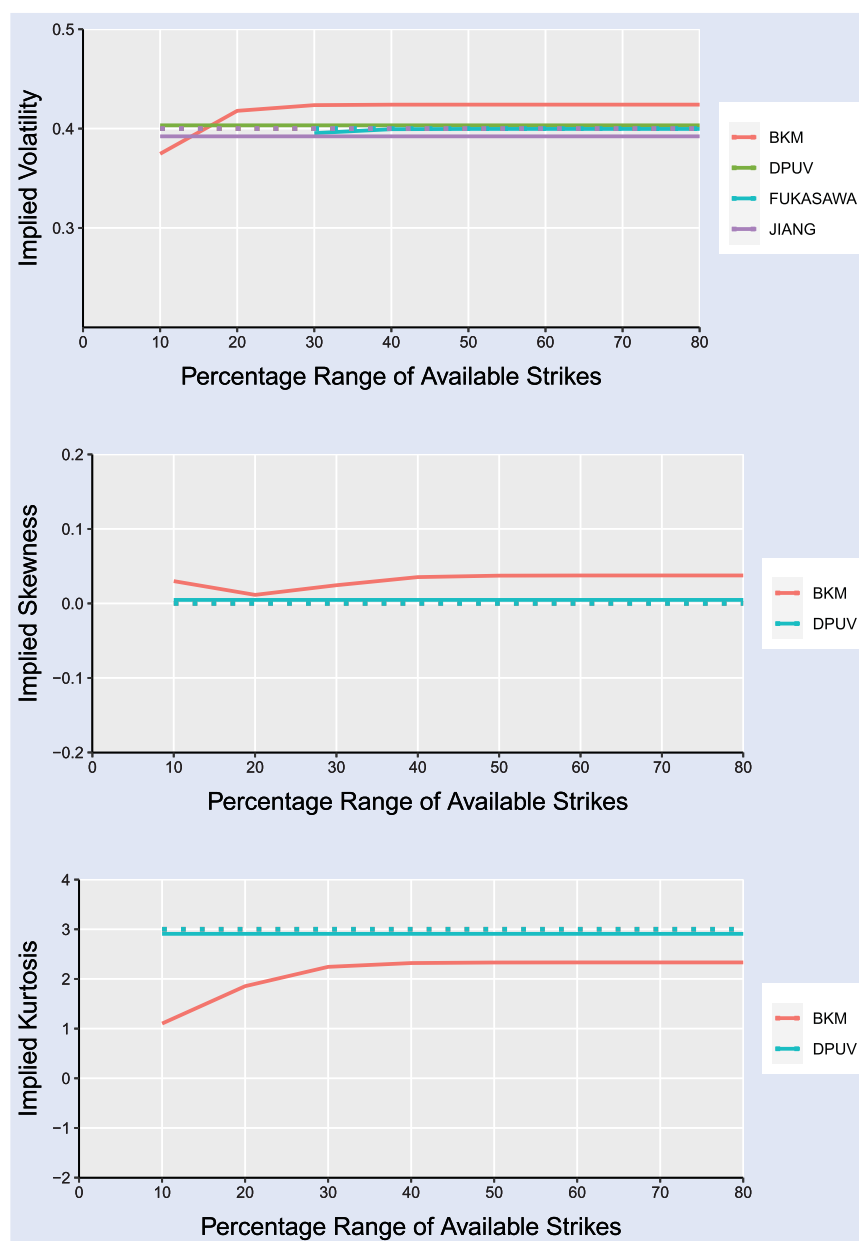


Figure 1. Simulation results – normally distributed returns with discrete strike prices. This figure illustrates the simulation results from calculating the Bakshi *et al.* (2003) implied moments using two different methodologies, the first the Conrad *et al.* (2013) approach and the other following DeMiguel *et al.* (2013). Model-free implied volatility estimates are also calculated following the methodology of Jiang and Tian (2005) and Fukasawa *et al.* (2011). We simulate Black Scholes option prices assuming the underlying spot price is \$50, the volatility is 40%, time to maturity is one month and the risk-free rate is 2%. We simulate prices for strike intervals of \$5 for various ranges. The horizontal axis represents the percentage range around the spot price that is observed for calculating the implied moments ranging from a 10% range (\$45–\$55) up to 80%. The true value for each of the implied moments is represented by the dotted line.

Our first practical test examines the impact of the discretization bias, the result of a firm only having limited, discrete strike prices versus continuous strike prices for a given tenor as required by BKM. Keeping all other parameters and the return generating process of the firm the same, we test how sensitive Dennis and Mayhew (2002) and DeMiguel *et al.* (2013) BKM implied moment estimates are to various ranges of moneyness with a \$5 interval between strikes. In figure 1, we show BKM implied volatility, implied skewness and implied kurtosis estimates using the methodologies of Dennis and Mayhew (2002) and DeMiguel *et al.* (2013), respectively. The ranges shown on the x-axis refer to the percentage band around the \$50 stock price that contains option

strike prices, i.e. a range of 20 would imply that option strikes between \$40 (\$50 – 20%) and \$60 (\$50 + 20%) exist and are used. As we increase the width of available strikes prices we have more information about the firm's implied return density, particularly the tail events, so a wider range yields a more accurate estimate of the BKM implied moments; especially higher moments. In unreported tests, we show the intuitive result that smaller strike price intervals limit the numerical integration error, and therefore reduce biases in BKM implied moments using both methodologies.

Figure 1 highlights our first concerns with the approach of Dennis and Mayhew (2002) followed by Conrad *et al.* (2013). None of the implied moment estimates using the approach



Figure 2. Simulation results – normally distributed returns with various spot prices This figure illustrates the simulation results from calculating the Bakshi *et al.* (2003) implied moments using two different methodologies, the first the Conrad *et al.* (2013) approach and the other following DeMiguel *et al.* (2013). Model-free implied volatility estimates are also calculated following the methodology of Jiang and Tian (2005) and Fukasawa *et al.* (2011). We simulate Black–Scholes option prices assuming the volatility is 40%, time to maturity is one month and the risk-free rate is 2%. We simulate prices for strike intervals of \$5 for a range from \$10 to \$90. The horizontal axis represents the spot price of the underlying asset and the three graphs show the estimated values of implied volatility, implied skewness and implied kurtosis, respectively. The true value for each of the implied moments is represented by the dotted line.

converge to their true value, irrespective of the range of option strike prices observed. When examining implied volatility, the Dennis and Mayhew (2002) approach is consistently biased upwards once the range of options observed is greater than roughly 20% and the estimate stabilizes once there is a range of 30% or more. However, when the range of options observed is below roughly 20% there is a downward bias in implied volatility. We attribute this to a lack of information to properly estimate implied volatility when too few option strike prices are available. In unreported tests, we confirm no bias in implied volatility using the approach of Dennis and Mayhew (2002) when there are strike prices at a 1-cent intervals

and there is a range of moneyness of plus and minus 40%. Similar results hold for \$1 strike intervals with the same range of moneyness. These results contrast greatly with those following the DeMiguel *et al.* (2013) approach which converges closely to the firm's actual implied volatility in each case, even when there is only a 10% range of moneyness observed and \$5 strike intervals.

The model-free implied volatility measures of Jiang and Tian (2005) and Fukasawa *et al.* (2011) fare well across the differing ranges of moneyness reported in figure 1. Regardless of the range of moneyness, the methodology of Jiang and Tian (2005) produces a very slightly downward biased

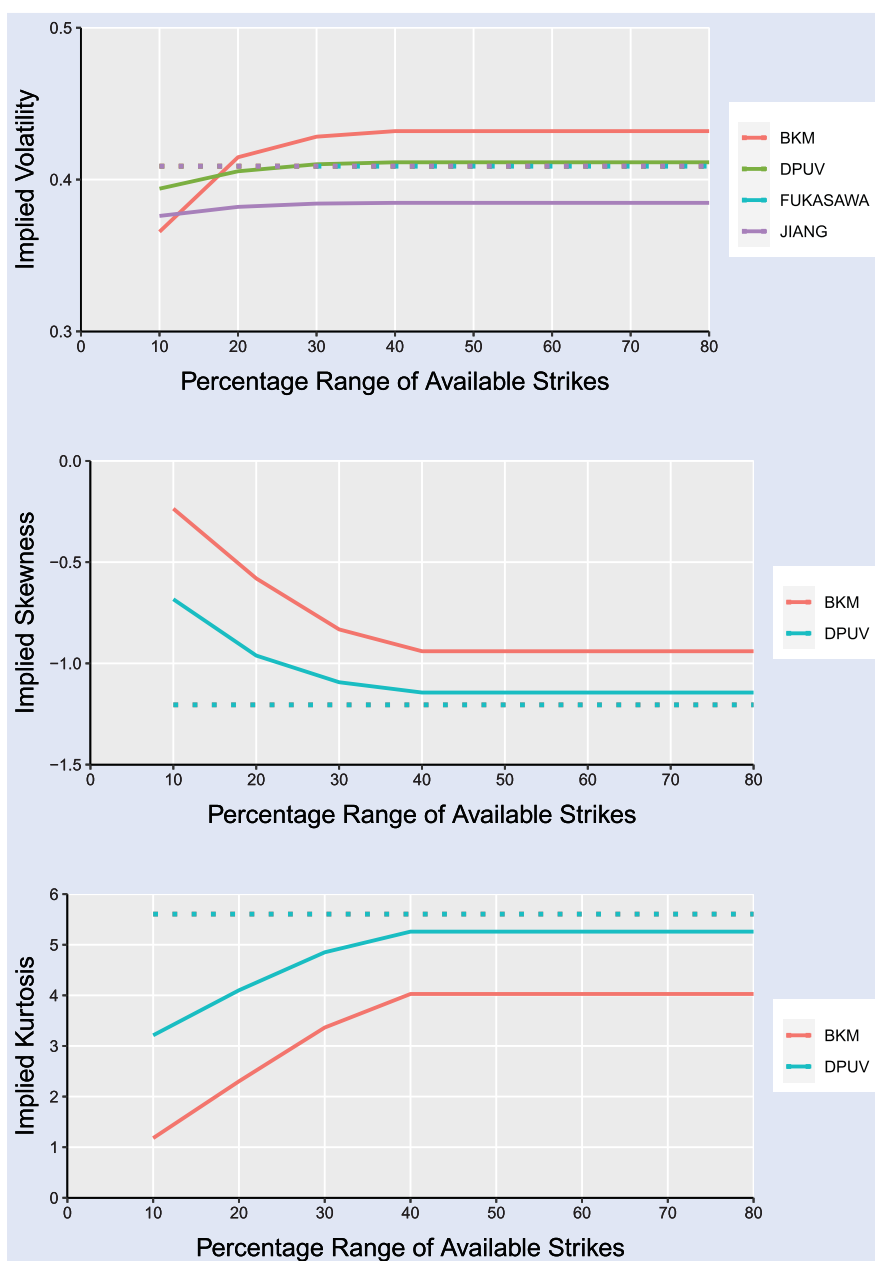


Figure 3. Simulation results – non-normally distributed returns with discrete strike prices This figure illustrates the simulation results from calculating the Bakshi *et al.* (2003) implied moments using two different methodologies, the first the Conrad *et al.* (2013) approach and the other following DeMiguel *et al.* (2013). Model-free implied volatility estimates are also calculated following the methodology of Jiang and Tian (2005) and Fukasawa *et al.* (2011). We simulate Heston stochastic volatility option prices assuming the underlying spot price is \$50, long-run volatility is 40% (θ or long-run variance is .16), time to maturity is one month, the risk-free rate is 2%, κ is 6.21, σ_v is 1.83, and ρ is $-.7$. We simulate prices for strike intervals of \$5 for various ranges. The horizontal axis represents the percentage range around the spot price that is observed for calculating the implied moments ranging from a 10% range (\$45–\$55) up to 80% range (\$10–\$90). The true value for each of the implied moments is represented by the dotted line.

estimate of IV (about 39.9% instead of 40.0% actual volatility of the simulated firm) and the methodology of Fukasawa *et al.* (2011) produces perfectly accurate estimates once the range of available strikes is 40% or greater. It should be noted that the methodology of Fukasawa *et al.* (2011) implicitly requires at least three out-of-the-money puts and at least three out-of-the-money calls, which is why results are only reported at and above a range of moneyness of 30% or greater (at a range of available strikes of 30% with a spot price of \$50 there exists three OTM puts, specifically a \$35 strike, \$40

strike, and a \$45 strike, and three OTM calls; specifically a \$55 strike, \$60 strike, and \$65 strike).

When examining implied skewness, a similar story holds in regards to the BKM estimates in which the Dennis and Mayhew (2002) approach creates a consistent upwards bias for all ranges of moneyness. The positive bias in implied skew stabilizes once there is a 40% range of moneyness. Again, in sharp contrast, the DeMiguel *et al.* (2013) approach is robust to all ranges of moneyness as it consistently provides precise and unbiased estimates of implied skewness.



Figure 4. Simulation results – non-normally distributed returns with various spot prices. This figure illustrates the simulation results from calculating the Bakshi *et al.* (2003) implied moments using two different methodologies, the first the Conrad *et al.* (2013) approach and the other following DeMiguel *et al.* (2013). Model-free implied volatility estimates are also calculated following the methodology of Jiang and Tian (2005) and Fukasawa *et al.* (2011). We simulate Heston stochastic volatility option prices assuming that long-run volatility is 40% (θ or long-run variance is .16), time to maturity is one month, the risk-free rate is 2%, κ is 6.21, σ_v is 1.83, and ρ is $- .7$. We simulate prices at \$5 strike intervals from \$10 to \$90. The horizontal axis represents the spot price of the underlying asset and the three graphs show the estimated values of implied volatility, implied skewness and implied kurtosis, respectively. The true value for each of the implied moments is represented by the dotted line.

Finally, while the Dennis and Mayhew (2002) approach demonstrates biases for both implied volatility and implied skewness estimates in the case of normally distributed firm returns, the magnitude of these biases is arguably not severe. However, the resulting negative bias in the implied kurtosis estimate is much larger in magnitude. Again, in our reported tests with \$5 strike intervals, this bias stabilizes once there is a 40% range of moneyness. Yet again, the DeMiguel *et al.* (2013) approach is robust to all ranges of moneyness, consistently providing precise estimates of implied kurtosis.

In summary, our first test of a firm with normally distributed returns and reasonably spaced, discrete strikes suggests that the DeMiguel *et al.* (2013) approach is incredibly robust to the discretization bias ; however, the Dennis and Mayhew (2002) approach has large shortcomings. Across the three moments, with a \$5 strike interval, it appears that a range of moneyness of at least plus or minus 40% is required to ensure that BKM implied moment estimates following the Dennis and Mayhew (2002) estimation methodology converge (however, the estimates still exhibit convergence to *biased* values for all three moments). Thus, a firm with a stock price of \$50 and the

parameters chosen, would need actively traded options from \$30, \$35, . . . , \$70 to get even a biased, but stable, measure of BKM implied moments following the technique of Dennis and Mayhew (2002). This is a much more active option series than the typical, optionable firm has for most tenors. Therefore, estimates of implied moments from Conrad *et al.* (2013) and others following Dennis and Mayhew (2002) are likely to be significantly biased for many firms from the *discretization bias* alone.

Our second test evaluates the extent to which each methodology is exposed to the ATM bias. Keep in mind, we postulate that this bias exists using the Dennis and Mayhew (2002) estimation methodology whenever an *exactly* at-the-money strike is lacking. To test this, we calculate implied moment estimates using each methodology keeping all parameters the same as the previous set of tests, with the exception that the firm's stock price is now flexible and the range of available strikes is fixed at 80%. In this set of tests we again use \$5 strike intervals, but now allow the firm's stock price to vary between \$45.01 and \$54.99 instead of being held fixed at \$50.00.

Figure 2 illustrates the results from this test of the ATM bias. As expected, the DeMiguel *et al.* (2013) approach is robust to all values of the underlying as the interpolation process ensures that there is always a true or synthetically created ATM option. In contrast, the Dennis and Mayhew (2002) approach followed by Conrad *et al.* (2013) is extremely sensitive to the ATM bias, even for minimal differences between the stock price and nearest strike (even 1-cent differences between the firm's stock price and the nearest strike price result in large biases).

Using the Dennis and Mayhew (2002) approach, there is a consistent and negative bias in implied volatility that is most extreme when the distance between the underlying asset and nearest strike price is maximized (i.e. the underlying asset is at the exact midpoint between the two nearest option strike prices, in our simulations, this occurs when the stock price is at \$47.50 or \$52.50). This contrasts with the slight upward bias in implied volatility with a \$5 strike interval when there is a true ATM option. The magnitude of the ATM bias' effect on implied volatility is substantial with estimates as low as 0.3, much lower than the true value of 0.4. This result turns out to actually be quite intuitive. When there is no at-the-money option, we are unable to fully represent the integrals in equation (5). Instead of the integration of OTM calls being over the range $[S(t), \infty)$ it will be over the range $[S(t) + \Delta, \infty)$ where Δ is the distance between the current spot price and the closest strike greater than $S(t)$. A similar argument holds for the second term of the equation. The part of the integral that is missing for both terms in the equation is positive, resulting in an estimate that is below the true value. Even more concerning, using the Dennis and Mayhew (2002) approach this bias still exists and is significant in magnitude even when the nearest strike price is only 1-cent from the underlying stock price. It is only erased when the underlying stock price is *exactly* equal to a strike price. This is a significant shortcoming to the suitability of the Dennis and Mayhew (2002) approach and causes any implications based on implied volatility following the approach to almost always be biased.

When using the Dennis and Mayhew (2002) approach and examining the effect of the ATM bias on implied skewness a different pattern presents. The bias in implied skewness is positive when the spot price is just below the nearest to the money strike, and becomes negative as the spot price is just above the nearest to the money strike. With \$5 strike intervals and a \$50 strike, when the spot is \$49.99 there is a large positive bias and when the spot is \$50.01 there is a large negative bias of the same magnitude. This is intuitive when we look at equation (6). When the spot is close to, but just above the nearest observed strike price the first term has a large negative bias while the second term only has a small negative bias resulting in a much smaller value for the cubic contract. When the spot is close to, but just below the nearest observed strike price we get the opposite effect resulting in a much larger value for the cubic contract; hence the positive bias.

The ATM bias in the estimate of implied kurtosis following Dennis and Mayhew (2002) occurs primarily as a result of the biases in the earlier moment estimation. Intuitively, the price of the quartic contract is going to be more greatly determined by far OTM options as opposed to near to the money options. As such, the bias in the implied kurtosis estimate is driven by the negative bias in the quartic contract. This causes the denominator of equation (11) to be smaller and the Dennis and Mayhew (2002) estimate of implied kurtosis to be extremely inflated when the firm's stock price is roughly midway between the nearest two strikes.

As expected, since the approach of DeMiguel *et al.* (2013) guarantees the existence of an *exactly* at-the-money strike, it suffers from no ATM bias. Implied volatility, implied skewness, and implied kurtosis estimates using the approach of DeMiguel *et al.* (2013) are robust whether or not the firm's stock price is at-the-money, at the midpoint between the nearest two strikes, or anywhere else in between.

Similarly, the model-free implied volatility approaches of Jiang and Tian (2005) and Fukasawa *et al.* (2011) perform very well in accurately extracting implied volatility when the firm's stock price is not at or near a strike price. This is not surprising as both methodologies involve, as one of their several steps, a cubic spline or polynomial interpolation of Black and Scholes (1973) IVs of existing options (thus, similar to DeMiguel *et al.* (2013), both methodologies guarantee the existence of a continuum of synthetically created put and call prices). Again, the methodology of Jiang and Tian (2005) exhibits an extremely slight downward bias in the IV estimate regardless of the spot price of the underlying, whereas the methodology of Fukasawa *et al.* (2011) exhibits perfectly accurate IV estimates regardless of the underlying spot price.

In summary, we show that in cases of log-normally distributed stock returns and constant variance the Dennis and Mayhew (2002) methodology followed by Conrad *et al.* (2013) suffers from problematic biases which drastically effect the estimates of BKM implied volatility, skewness and kurtosis. On the other hand, the DeMiguel *et al.* (2013) approach is robust to such limitations and potential biases. However, there is still the question as to how the two approaches perform in the most extant case; when returns are non-normal. In fact, the median firm exhibits negative return skewness and excess return kurtosis. Moreover, the distribution of firms' measures of skew and kurtosis within the

cross-section is staggeringly wide. This general pattern also occurs in cross-sectional estimates of BKM implied moments. Thus, determining the biases of each approach in estimating BKM implied moments for firms with negatively skewed, leptokurtotic returns is paramount.

Following the procedure, we describe in section 2, we use the Heston (1993) stochastic volatility model to create a reasonable return generating process for a typical firm in the cross-section. Using Monte Carlo simulation, we are able to numerically estimate this hypothetical firm's actual return distribution by taking the expected values of the firm's realized moments over the simulated return paths. We simulate 1 million return paths and find that 1 million paths are more than ample to ensure convergence of our numerical estimation to the actual moments.

We now examine how non-normally distributed returns bias BKM implied moment estimation. In doing so, it's important to recognize that the bias due to non-normality is essentially the synthesis of the ATM Bias and Discretization bias, which becomes amplified in the case of a firm with non-normally distributed returns. For mathematical reasons, it turns out that the ATM bias and Discretization Bias are actually compounded in the non-normal case. Of course, this implies that for most firms in the cross-section the biases in the estimates of BKM implied moments using the approach of Dennis and Mayhew (2002) are greater than reported in our prior panel of tests (which assume normally distributed returns).

The hypothetical firm that we simulate using Heston's (1993) model has a stock price of \$50 and a time to maturity of its options, τ , of 1-month. This allows one to somewhat fairly compare the magnitudes of the ATM bias and the Discretization bias in the Black–Scholes case above to the case of non-normally distributed returns we detail herein. The firm's parameters within Heston's SV model are as follows: κ is 6.21, θ is 0.16, σ_v is 1.83, ρ is -0.7, and the risk-free rate, r , is 0.02.

The results of our simulations which detail the impact of the discretization bias in the case of the above firm with non-normal returns are reported in figure 3. Following our prior methodology we set strike prices at \$5 intervals, and simulate ranges of moneyness around the firm's stock price of 10% to 80%. Using the Dennis and Mayhew (2002) estimation approach, we find that the biases in all three implied moments are in the same direction and follow a convergence pattern similar to that found in the analogous test with normally distributed returns (results reported in figure 1). As expected, the magnitude of each bias is now significantly larger due to the left-skewness and excess kurtosis introduced into the firm's actual return distribution. In contrast to the approach of Dennis and Mayhew (2002), the DeMiguel *et al.* (2013) approach shows perfect convergence of the implied volatility estimate to the actual value at a moneyness range of approximately 30%. However, the approach does now exhibit a small positive bias in implied skewness, and a small negative bias in implied kurtosis. Both estimates exhibit convergence to slightly biased implied moment estimates at a moneyness range of approximately 40%. While BKM implied skew and implied kurtosis estimates using the DeMiguel *et al.* (2013) approach are not perfect in the non-normal case, they are

drastically more accurate than estimates provided using the alternative approach of Dennis and Mayhew (2002).

The model-free implied volatility approach of Fukasawa *et al.* (2011) performs extremely well in accurately extracting implied volatility in the case of non-normality, with perfectly accurate estimates across all calculable ranges of moneyness. The Jiang and Tian (2005) approach again exhibits a consistent downward bias in IV that seems to be increased due to the non-normality introduced. We do not speculate as to why this downward bias is present, nor guarantee that it is not a result of our implementation of the Jiang and Tian (2005) methodology, but we notice that the bias seems to mirror the upward bias in implied volatility found within the same scenario when using the Dennis and Mayhew (2002) approach to calculate BKM implied volatility.

Our simulation results which detail the impact of the ATM bias in the case of the hypothetical firm with non-normal returns are reported in figure 4. Again, following our prior methodology, we set strike prices at \$5 intervals, set the range of moneyness around the firm's stock price to 80%, and simulate starting stock prices from \$45.01 to \$54.99 in 1-cent intervals. We find that using the approach of Dennis and Mayhew (2002) the pattern of bias in BKM implied volatility is very similar in shape, but slightly smaller in magnitude to the analogous case with normally distributed returns. However, the size of the implied skewness and implied kurtosis biases are much larger with left-skewness and excess kurtosis introduced into returns. In addition, while the shape of the implied volatility and implied kurtosis estimates across varying spot stock prices is similar in both the normal and non-normal cases, the effect on implied skew differs in the non-normal case. After left-skewness in firm returns is introduced, the sign change in implied skewness using the Dennis and Mayhew (2002) approach disappears. Instead, a consistent downward bias in implied skew, which gets very large at the midpoint between the nearest strikes, is examined in all cases in which an exactly ATM option does not exist. In the case that a true ATM option does exist, the bias becomes positive and is smaller in magnitude. Unsurprisingly, the opposite but analogous pattern results when right-skewness is introduced. This contrasts with the prior case of normally distributed firm returns in which the sign of the bias in implied skew using the approach of Dennis and Mayhew (2002) depends on whether the firm's stock price is just above or just below the nearest-to-the-money strike. In totality, biases in implied skewness and implied kurtosis under the Dennis and Mayhew (2002) method of estimation are significantly compounded when introducing skewness or excess kurtosis into returns.

We next examine the ATM bias with non-normal firm returns using the DeMiguel *et al.* (2013) approach. In this case, we find almost perfect convergence to true moments occurs for all three of the implied moment estimates. Very minimal biases exist at various stock prices; however, their magnitude is extremely minimal and no clearly discernible pattern in these slight biases exists. We posture that these very slight biases are the result of the DeMiguel *et al.* (2013) method's dependence of imputation of IVs using the Black and Scholes (1973) model. While this is certainly standard

and correct, these IVs assume normality as a key assumption of Black-Scholes options pricing is that of normally distributed asset returns (or analogously, lognormally distributed asset prices) and thus the resulting synthetic option prices created from the Black and Scholes (1973) approach are also slightly incorrect due to the firm's left-skewness and excess kurtosis. Still, following the approach of DeMiguel *et al.* (2013) and using these synthetically created option prices to implement the BKM methodology, and then fitting a cubic spline to admittedly imperfect price estimates based on Black-Scholes IVs results in *far smaller* implied moment biases than using the Dennis and Mayhew (2002) approach which suffers from large gaps between strikes and, in almost all cases, no exactly ATM option. We prove that the advantage of DeMiguel *et al.* (2013) approach holds in cases in which the underlying asset's return generating process is normal or non-normal. If anything, the relative advantages of the DeMiguel *et al.* (2013) approach are only magnified in the non-normal case.

Finally, when examining the model-free implied volatility estimates of Jiang and Tian (2005) and Fukasawa *et al.* (2011) under the case of non-normality with varying spot prices, we again see that the methodology of Fukasawa *et al.* (2011) provides perfect IV estimates regardless of the spot price, whereas Jiang and Tian's methodology provides stable IV estimates, with a very slight downward bias regardless of the underlying spot price.

In summary, our simulation tests raise serious concerns about the accuracy of the BKM implied moment estimates using the Dennis and Mayhew (2002) approach applied in leading studies including Conrad *et al.* (2013). In the realistic cases we examine for firms with normal and non-normal returns and varying options microstructures, the method exhibits large biases. Using the Dennis and Mayhew (2002) approach, these large biases could all but ruin any effort to consistently relate firms' realized returns (or anything else for that matter) to firms' implied moments. In contrast, the DeMiguel *et al.* (2013) approach works extremely well in the case of normally distributed returns. While the DeMiguel *et al.* (2013) method does exhibit some minor biases in the non-normal case, the biases are very small in magnitude compared to the Dennis and Mayhew (2002) approach. Based on this, we are able to confidently recommend the use of the DeMiguel *et al.* (2013) approach in favor of the Dennis and Mayhew (2002) approach.

In addition, we recommend either the use of the Fukasawa *et al.* (2011) model-free implied volatility methodology or the BKM methodology using the DeMiguel *et al.* (2013) approach if one is only cornered with an IV estimate. Both methodologies are exceedingly accurate across all cases, with minimal or even zero bias in estimating IV. Of course, the use of the DeMiguel *et al.* (2013) approach to extract BKM implied moments would be necessary for those looking to extract the entire implied density, including the third and fourth moment estimates.

Finally, we would like to add the caveat that even the DeMiguel *et al.* (2013) approach may suffer from economically meaningful biases when a sufficient range of strikes is lacking (a range of moneyness of 40% around the firm's stock price is sufficient in all cases we test).

Given these results, we now examine the implications of BKM implied moment estimation in the realm of empirical asset pricing. We show that the inferences of a leading work which examines BKM implied moments and the cross-section of stock returns using the methodology of Dennis and Mayhew (2002) differ once the superior methodology of DeMiguel *et al.* (2013) is applied.

4.2. Implied moments and the cross-section of returns

Now that we have identified the limitations and relative benefits of the two leading approaches to estimate BKM implied moments we demonstrate how these approaches may lead to different inferences in practice. Specifically, we demonstrate that the past inferences of Conrad *et al.* (2013) regarding the relation between option-implied moments and the cross-section of stock returns are related to the biases in the Conrad *et al.* (2013) estimation of BKM implied moments (which follows the methodology of Dennis and Mayhew 2002). We start our analysis with the daily options dataset from 2002 to 2016 and apply the data filters outlined in section 3. We then calculate the BKM implied moments using both the Conrad *et al.* (2013) and DeMiguel *et al.* (2013) estimation methodologies. We show that the DeMiguel *et al.* (2013) approach leads to different inferences as to how the implied moments relate to the cross-section of stock returns. Since we definitively prove that the DeMiguel *et al.* (2013) approach is much more accurate and exudes less bias and inconsistency than the approach of Dennis and Mayhew (2002) followed by Conrad *et al.* (2013); we argue that our new inferences regarding firms' option-implied moments and the cross-section of stock returns using the methodology of DeMiguel *et al.* (2013) should be considered in favor of the results reported using the approach of Dennis and Mayhew (2002).

Table 1 shows the summary statistics of our estimates of implied volatility, implied skewness and implied kurtosis using the two approaches. From the start, it is evident that the two approaches provide very different option-implied moments. When examining implied volatility, the Conrad *et al.* (2013) approach has an annual median in the mid 20% to low 30% range. In contrast, the DeMiguel *et al.* (2013) approach has a much higher median in the mid 30% to 50% range. Consistent with our simulation results, the BKM implied volatility estimates using the Conrad *et al.* (2013) approach are much lower (biased downward) than the same estimates using the DeMiguel *et al.* (2013) approach. This is due to the ATM bias, which causes the Conrad *et al.* (2013) BKM implied volatility estimates to be biased downward for any firm in which the stock price is not exactly at-the-money; a condition which would almost never happen (and even then, only for a brief moment in time) in practice. It is well known that the annualized realized volatility of the typical (median) firm within the cross-section is roughly 40%, consistent with our DeMiguel *et al.* (2013) estimates of implied volatility. While the magnitudes and dispersion of Conrad *et al.* (2013) and DeMiguel *et al.* (2013) implied volatility estimates differ significantly; general patterns over the time-series are relatively similar as implied volatility increases from 2005 to 2007 using both approaches, reaching maximal levels during

Table 1. Descriptive statistics: risk-neutral moments.

Model	Year	Volatility			Skewness			Kurtosis		
		P5	P50	P95	P5	P50	P95	P5	P50	P95
CDG	2002	16.88	30.77	58.97	-2.47	-1.04	-0.15	2.01	4.20	13.41
	2003	14.55	25.53	48.13	-2.63	-1.09	-0.20	2.10	4.60	14.84
	2004	12.44	21.90	41.18	-2.64	-1.08	-0.18	2.26	4.89	14.85
	2005	11.46	19.97	36.71	-2.99	-1.17	-0.27	2.45	5.54	17.40
	2006	11.08	21.10	38.63	-3.17	-1.19	-0.28	2.46	5.71	18.98
	2007	12.18	23.05	44.10	-2.94	-1.18	-0.31	2.42	5.29	16.10
	2008	18.32	33.88	71.46	-2.36	-1.06	-0.22	1.91	4.13	11.95
	2009	19.38	35.94	67.43	-2.20	-0.98	-0.22	1.82	3.88	12.70
	2010	14.89	26.80	46.35	-2.47	-0.97	-0.14	1.91	4.38	15.69
	2011	14.57	27.86	49.60	-2.61	-0.97	-0.13	1.84	4.35	16.97
	2012	12.88	26.10	47.25	-2.93	-1.01	-0.16	1.83	4.54	19.62
	2013	11.48	22.69	43.78	-2.42	-0.89	-0.07	1.94	4.72	17.23
	2014	10.85	22.31	48.36	-2.12	-0.78	0.06	1.96	4.61	15.45
	2015	11.53	23.53	51.56	-2.31	-0.86	-0.01	1.95	4.71	16.98
	2016	12.81	26.93	59.68	-2.78	-1.05	-0.12	1.90	4.84	20.20
DPUV	2002	29.63	51.16	89.61	-1.01	-0.47	0.05	1.75	2.90	3.83
	2003	25.63	41.68	72.61	-1.05	-0.53	-0.04	1.90	3.00	3.84
	2004	21.83	35.39	63.16	-1.11	-0.49	-0.11	2.18	3.05	3.95
	2005	20.36	32.34	55.51	-1.17	-0.51	-0.15	2.37	3.11	4.18
	2006	20.46	34.70	58.24	-1.11	-0.46	-0.13	2.27	3.05	4.07
	2007	21.90	36.44	63.72	-1.08	-0.49	-0.11	2.10	3.04	3.94
	2008	30.47	53.03	101.21	-0.91	-0.44	0.09	1.65	2.88	3.57
	2009	33.26	55.70	92.45	-0.92	-0.48	0.06	1.62	2.88	3.58
	2010	27.27	42.35	66.71	-1.09	-0.56	-0.14	2.14	3.09	4.14
	2011	26.35	43.30	72.24	-1.21	-0.58	-0.11	2.17	3.14	4.52
	2012	24.15	40.25	67.76	-1.38	-0.63	-0.12	2.17	3.21	5.01
	2013	21.18	33.89	63.80	-1.19	-0.54	-0.09	2.38	3.22	4.64
	2014	19.71	33.12	71.58	-1.13	-0.45	0.03	2.46	3.22	4.74
	2015	20.99	34.74	76.51	-1.24	-0.51	-0.01	2.42	3.28	5.06
	2016	23.81	40.09	89.02	-1.39	-0.61	-0.03	2.07	3.31	5.45

Note: This table shows the distribution of risk-neutral moments extracted from options and therefore estimates are conditional for a given horizon. We follow the approach in Bakshi *et al.* (2003) using data on out-of-the-money options to calculate implied moments. We restrict the sample to all out of the money puts and calls with closing prices in excess of \$0.50 and not expiring within a week. We also exclude options that violate put-call parity restrictions and have less than 10 observations in a month. We follow two applications of this procedure. In the top portion of the table, we follow the methodology applied by Conrad *et al.* (2013) which follow Dennis and Mayhew (2002). In the bottom portion of the table, we follow the methodology of DeMiguel *et al.* (2013). The Conrad *et al.* (2013) approach requires an extra data filter such that all option series that survive the above filters have at least two OTM puts and two OTM calls. Following Dennis and Mayhew (2002), within the Conrad *et al.* (2013) methodology when there is an uneven number of OTM puts and calls, we truncate the put and call strikes used to ensure symmetry in strikes around the underlying stock price and maintain an even number of put and call observations. The Conrad, Dittmar and Ghysels sample consists of 8,801,731 firm-day-expiration observations and the DeMiguel, Plyakha, Uppal and Vilkov sample (2013) sample consists of 8,849,197 firm-day-expiration observations over the time period of 2002 to 2016. Entries in the table are the 5th percentile (P5), median (P50), and 95th (P95) percentile of risk-neutral volatility, skewness and kurtosis estimates across securities by year.

the 2008–2009 financial crisis and then declining in the ensuing bull market from 2009 to 2016.

When examining the implied skewness estimates the approach of Conrad *et al.* (2013) indicates much more left-skewness for the median stock. While, the approach of DeMiguel *et al.* (2013) also indicates minor left-skewness for the median stock (a universally accepted result), the amount of left-skewness is considerably dampened. Again, this difference in the implied moment estimates can primarily be attributed to the ATM bias and is completely consistent with our simulation tests. In the non-normal case for a firm with slight left-skewness in return, we demonstrate within figure 4 that the Conrad *et al.* (2013) estimate of the firm's skewness

will be biased downward in all cases that the firm's underlying stock is not exactly at-the-money; and that the magnitude of this bias is maximized when the firm's stock is at the exact midpoint between the two nearest strikes. This will lead to a significant downward bias in BKM implied skew as shown in table 1. In addition, the Conrad *et al.* (2013) approach exhibits a much wider distribution of implied skew estimates with a much lower 5th percentile of implied skew estimates than the DeMiguel *et al.* (2013) approach. This is again consistent with our simulation results and attributable to the distance between the firm's stock price and the nearest options strike prices, rather than having anything to do with a firm's actual or correctly estimated skew.

Finally, when examining implied kurtosis, the Conrad *et al.* (2013) approach indicates a considerable amount of implied leptokurtosis for the typical (median) stock, whereas the DeMiguel *et al.* (2013) approach indicates that the typical stock has an implied kurtosis of just above 3, nearly in line with what would be expected if stock's returns perfectly followed a normal distribution. In addition, the Conrad *et al.* (2013) approach results in much more extreme implied kurtosis estimates than the DeMiguel *et al.* (2013) approach, particularly at the 95th percentile level as Conrad *et al.* (2013) 95th percentile implied kurtosis is at absurd levels of 15 to 20 in various years within the data. Again, the intuition of this completely matches our simulation test within figure 4, in which we demonstrate that the implied kurtosis of a typical firm within the cross-section with an actual kurtosis of 5 can easily have Conrad *et al.* (2013) implied kurtosis estimates in the 10 to 15 range if the firm's stock price is nearly midway between the two closest strikes for the observed tenor.

As noted by BKM, daily estimates are very noisy. Therefore, following Conrad *et al.* (2013) we average the estimates over a calendar quarter for 3 maturities; 1) 3 months, 2) 6 months and 3) 12 months. Each day, for each underlying asset, we assign the nearest to 90 day expiration as the 3 month contract, nearest to 180 days as the 6 month, and nearest to 360 day as the 12 month. Then, we average the daily estimate of each firm's implied moments over the calendar quarter to get a quarterly estimate of the firm's implied moments. However, we first examine the correlation between daily estimates of the two approaches to estimate implied moments in table 2.

When examining the relationship in implied moment estimates between the two approaches we find that higher moment estimates exhibit shockingly low correlations. Using daily data, while the correlation between Conrad *et al.* (2013) and DeMiguel *et al.* (2013) implied volatility estimates is +0.89, the correlation between the implied skewness and implied kurtosis estimates is only +0.55 and +0.45, respectively. This is again consistent with our simulation exercises as the size of the biases in the Conrad *et al.* (2013) approach expands for higher moment estimates. The relationship between the implied moments under each approach also differs. For example, the Conrad *et al.* (2013) approach

shows that implied volatility and implied skewness have little relationship with a correlation of +0.23, while implied volatility and implied kurtosis have a -0.41 correlation. Using the DeMiguel *et al.* (2013) approach, implied volatility and implied skewness exhibit a stronger correlation of +0.40 whereas implied volatility and implied kurtosis exhibit a stronger negative correlation of -0.48.

In our final tests, we separately apply each approach and examine the cross-section of stock returns in relation to the resulting implied moment estimates. Firms are sorted into tercile portfolios based on their implied moment estimate under each approach. Following the standard definition of a tercile portfolio sort within the empirical asset pricing literature, the bottom tercile portfolio holds firms below the 30% of the implied moment estimate, the middle tercile portfolio holds firms from the 30% to the 70% of the estimate, and the top tercile portfolio holds firms above the 70% of the estimate. Thus, 40% of firms fall in the middle tercile portfolio in each sort. Portfolio assignments are then held constant for each subsequent calendar quarter. Table 3 shows the results for a single sort on the implied moments for options with 3-months (90 days) to expiration and table 4 shows the results for a single sort on the implied moments for options with 12-months (360 days) to expiration. Similarly, tables 5 and 6 show the results for portfolios double sorted (independently) on implied volatility and either implied skewness or implied kurtosis; with implied moments again measured using options with 3-months (table 5) or 12-months (table 6) until maturity.

First, we briefly report the results of the 3-month maturity single sorted portfolios shown in table 3. We note that all of our results regarding portfolio returns are rounded to 10 bps or .1%. Using the differing methodologies of Conrad *et al.* (2013) and DeMiguel *et al.* (2013), mean cumulative quarterly (3-month) returns and mean quarterly (3-month) characteristic adjusted returns vary somewhat when examining tercile portfolios sorted on implied volatility, implied skewness and implied kurtosis. A long-short, zero net cost high implied volatility (tercile 3) minus low implied volatility (tercile 1) portfolio loses -30 bp per quarter using the Conrad *et al.* (2013) approach to estimate moments, whereas the same portfolio loses a very similar -20 bp per quarter using

Table 2. Implied moment correlations.

Correlation Matrix: Daily Implied Moments					
	CDG_Vol	CDG_Skew	CDG_Kurt	DPUV_Vol	DPUV_Skew
CDG_Skew	0.230				
CDG_Kurt	-0.408	-0.718			
DPUV_Vol	0.893	0.136	-0.239		
DPUV_Skew	0.315	0.546	-0.413	0.397	
DPUV_Kurt	-0.419	-0.295	0.445	-0.480	-0.737

Note: This table reports the correlation matrix for the two different methodologies to calculate the risk-neutral moments. We follow the procedure in Bakshi *et al.* (2003) using data from out-of-the-money options. We follow two applications of this procedure, one following Conrad *et al.* (2013) and the other following DeMiguel *et al.* (2013). We require at least two OTM puts and two OTM calls with prices in excess of \$0.50 which are not expiring within 1 week. The CDG sample consists of 8,801,731 firm-day-expiration observations and the DPUV sample consists of 8,849,197 firm-day-expiration observations over the time period from 2002 to 2016. The correlation reported is a Pearson Correlation between implied moments at the firm-day-expiration level.

Table 3. Descriptive statistics: risk-neutral moment portfolios – 3 months.

	Return							
Tercile	Mean	Char-Ajd	Vol	Skew	Kurt	Beta	ln MV	B/M
Panel A: CDG Methodology								
Volatility								
1	0.025	0.004	0.167	− 1.132	9.266	0.948	7.247	0.499
2	0.019	− 0.003	0.239	− 0.984	7.048	1.185	6.888	0.551
3	0.021	0.000	0.354	− 0.870	5.301	1.540	6.624	0.653
diff	− 0.003	− 0.003	0.187	0.262	− 3.965	0.592	− 0.623	0.155
p value	0.563	0.493	0.000	0.000	0.000	0.000	0.000	0.000
Skewness								
1	0.019	− 0.002	0.232	− 1.533	10.434	1.195	7.030	0.542
2	0.024	0.003	0.254	− 0.949	6.404	1.217	6.914	0.563
3	0.021	− 0.001	0.269	− 0.514	4.980	1.251	6.806	0.595
diff	0.002	0.000	0.037	1.019	− 5.455	0.056	− 0.224	0.054
p value	0.762	0.940	0.000	0.000	0.000	0.000	0.000	0.000
Kurtosis								
1	0.021	0.000	0.293	− 0.631	3.774	1.297	6.769	0.643
2	0.022	0.000	0.253	− 0.986	6.395	1.226	6.914	0.549
3	0.021	0.000	0.209	− 1.369	11.668	1.137	7.068	0.512
diff	− 0.000	0.000	− 0.084	− 0.738	7.894	− 0.160	0.300	− 0.131
p value	0.993	0.943	0.000	0.000	0.000	0.000	0.000	0.000
Panel B: DPUV Methodology								
Volatility								
1	0.024	0.002	0.286	− 0.662	3.522	0.925	7.267	0.489
2	0.019	− 0.002	0.381	− 0.570	3.413	1.185	6.917	0.545
3	0.022	0.001	0.539	− 0.485	3.219	1.564	6.566	0.672
diff	− 0.002	− 0.001	0.253	0.177	− 0.303	0.639	− 0.701	0.183
p value	0.779	0.892	0.000	0.000	0.000	0.000	0.000	0.000
Skewness								
1	0.017	− 0.003	0.357	− 0.824	3.825	1.132	7.182	0.526
2	0.023	0.001	0.392	− 0.555	3.302	1.208	6.895	0.569
3	0.024	0.003	0.453	− 0.342	3.062	1.325	6.679	0.603
diff	0.007	0.006	0.096	0.482	− 0.764	0.193	− 0.503	0.077
p value	0.192	0.198	0.000	0.000	0.000	0.000	0.000	0.000
Kurtosis								
1	0.026	0.004	0.464	− 0.378	3.008	1.340	6.692	0.634
2	0.020	− 0.001	0.386	− 0.551	3.297	1.212	6.915	0.569
3	0.018	− 0.002	0.353	− 0.795	3.886	1.112	7.145	0.495
diff	− 0.008	− 0.006	− 0.111	− 0.417	0.878	− 0.228	0.453	− 0.139
p value	0.156	0.188	0.000	0.000	0.000	0.000	0.000	0.000

Note: Panels A and B above report the summary statistics of tercile portfolios sorted by BKM implied moments within each calendar quarter. Each quarter, we form equally weighted tercile portfolios of firms by each implied moment, holding the ranking constant for the subsequent calendar quarter. For each firm, the options expiration with a time-to-maturity closest to 3-months is used to estimate that firm's implied moments. We calculate the risk-neutral implied moments following equations in Bakshi *et al.* (2003) and in Panel A report the results using the Conrad *et al.* (2013) methodology and Panel B report results using the DeMiguel *et al.* (2013) methodology. The first column of each panel reports the mean cumulative quarterly (3-month) return and the second the mean characteristic adjusted return of the tercile moment portfolio over the quarterly periods using Fama and French (1993) 5 × 5 size and book-to-market portfolios to benchmark. The next three columns report empirical estimates of the portfolio's average annualized volatility, skewness and kurtosis over the quarterly (3-month) periods. The final three columns report the Beta, log of the market capitalization, and book-to-market ratio of the implied moment tercile portfolios.

the approach of DeMiguel *et al.* (2013). However, differences become more noticeable when examining portfolios sorted on implied skew or implied kurtosis. The long-short, zero net cost high skew minus low skew portfolio earns +20 bp per quarter using the Conrad *et al.* (2013) approach to estimate moments, whereas the same portfolio performs considerably better, earning +70 bp a quarter when using the DeMiguel *et al.* (2013) approach to estimate BKM implied moments. Finally, the high kurtosis minus low kurtosis portfolio earns +0 bp per quarter when using the Conrad *et al.* (2013)

approach to estimate implied moments, whereas the same portfolio loses −80 bp per quarter using the DeMiguel *et al.* (2013) approach. In addition, the differences in long-short portfolio returns sorted on the implied moments only become larger in magnitude when changing the tenor of the option used for estimation from 3-months to 12-months.

At the 12-month maturity reported in table 4, when using the Conrad *et al.* (2013) methodology, the long-short high implied volatility minus low implied volatility portfolio loses −30 bp per quarter, whereas the same portfolio actually earns

Table 4. Descriptive statistics: risk-neutral moment portfolios – 12 months.

	Return							
Tercile	Mean	Char-Ajd	Vol	Skew	Kurt	Beta	ln MV	B/M
Panel A: CDG Methodology								
Volatility								
1	0.020	− 0.001	0.175	− 1.803	9.748	0.797	7.288	0.483
2	0.026	0.006	0.250	− 1.441	6.942	1.108	7.028	0.577
3	0.017	− 0.006	0.375	− 1.243	5.005	1.571	6.715	0.710
diff	− 0.003	− 0.005	0.201	0.560	− 4.743	0.773	− 0.573	0.227
p value	0.630	0.341	0.000	0.000	0.000	0.000	0.000	0.000
Skewness								
1	0.012	− 0.008	0.223	− 2.186	10.957	1.007	7.208	0.560
2	0.022	0.000	0.270	− 1.433	6.477	1.166	7.032	0.578
3	0.030	0.009	0.301	− 0.869	4.408	1.284	6.789	0.631
diff	0.018	0.017	0.078	1.318	− 6.549	0.277	− 0.419	0.071
p value	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Kurtosis								
1	0.025	0.004	0.328	− 0.990	3.783	1.358	6.791	0.653
2	0.022	0.000	0.260	− 1.435	6.351	1.130	7.054	0.576
3	0.017	− 0.003	0.208	− 2.065	11.766	0.980	7.178	0.541
diff	− 0.008	− 0.008	− 0.120	− 1.075	7.982	− 0.378	0.388	− 0.111
p value	0.158	0.121	0.000	0.000	0.000	0.000	0.000	0.000
Panel B: DPUV Methodology								
Volatility								
1	0.018	− 0.002	0.284	− 0.936	3.719	0.762	7.370	0.465
2	0.023	0.002	0.380	− 0.705	3.282	1.109	7.039	0.567
3	0.023	0.000	0.541	− 0.447	2.675	1.614	6.609	0.745
diff	0.005	0.002	0.256	0.489	− 1.044	0.852	− 0.761	0.279
p value	0.378	0.656	0.000	0.000	0.000	0.000	0.000	0.000
Skewness								
1	0.013	− 0.007	0.318	− 1.056	3.877	0.890	7.385	0.522
2	0.018	− 0.004	0.386	− 0.691	3.168	1.123	7.011	0.586
3	0.036	0.013	0.499	− 0.346	2.666	1.466	6.631	0.662
diff	0.023	0.020	0.180	0.710	− 1.210	0.576	− 0.754	0.140
p value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Kurtosis								
1	0.028	0.005	0.520	− 0.393	2.592	1.533	6.628	0.715
2	0.022	0.001	0.377	− 0.677	3.183	1.091	7.016	0.567
3	0.015	− 0.005	0.309	− 1.029	3.937	0.863	7.384	0.494
diff	− 0.013	− 0.010	− 0.212	− 0.636	1.345	− 0.671	0.756	− 0.221
p value	0.033	0.047	0.000	0.000	0.000	0.000	0.000	0.000

Note: Panels A and B above report the summary statistics of tercile portfolios sorted by BKM implied moments within each calendar quarter. Each quarter, we form equally weighted tercile portfolios of firms by each implied moment, holding the ranking constant for the subsequent calendar quarter. For each firm, the options expiration with a time-to-maturity closest to 12-months is used to estimate that firm's implied moments. We then calculate the risk-neutral implied moments following equations in Bakshi *et al.* (2003) and in Panel A report the results using the Conrad *et al.* (2013) methodology and Panel B report results using the DeMiguel *et al.* (2013) methodology. The first column of each panel reports the mean cumulative quarterly (3-month) return and the second the mean characteristic adjusted return of the tercile moment portfolio over the quarterly periods using Fama and French (1993) 5 × 5 size and book-to-market portfolios to benchmark. The next three columns report empirical estimates of the portfolio's average annualized volatility, skewness and kurtosis over the quarterly (3-month) periods. The final three columns report the Beta, log of the market capitalization, and book-to-market ratio of the implied moment tercile portfolios.

+ 50 bp a quarter when using the DeMiguel *et al.* (2013) methodology. When examining portfolios sorted on implied skew, under the Conrad *et al.* (2013) methodology the long-short high implied skew minus low implied skew portfolio earns + 180 bp a quarter, whereas the same portfolio earns + 230 bp a quarter when formed using DeMiguel *et al.* (2013) methodology. Finally, the long-short high kurtosis minus low kurtosis portfolio loses − 80 bp a quarter when implied kurtosis is estimated using the Conrad *et al.* (2013) methodology,

but loses − 130 bp a quarter when implied kurtosis is estimated using the DeMiguel *et al.* (2013) approach.

In totality, results of tables 3 and 4 highlight a significant concern that the impact of implied moments on the cross-section of stock returns may be heavily dependent on the BKM estimation methodology employed. For example, using the 12-month estimation window, our results indicate that when using the approach of DeMiguel *et al.* (2013), which we prove to be more accurate under all circumstances

Table 5. Risk-neutral moment independent double-sorted portfolios – 3 months.

Vol	Return					Char-Adj Return				
	1	2	3	diff	<i>p</i> value	1	2	3	diff	<i>p</i> value
Panel A: CDG Methodology										
Skewness										
1	0.032	0.021	−0.013	−0.045	0.000	0.006	−0.003	−0.014	−0.020	0.063
2	0.024	0.015	0.035	0.011	0.224	0.003	−0.003	0.009	0.006	0.462
3	0.011	0.021	0.025	0.013	0.196	0.000	−0.002	−0.001	−0.001	0.868
diff	−0.020	−0.000	0.038			−0.006	0.001	0.013		
<i>p</i> value	0.005	0.994	0.009			0.354	0.833	0.303		
Kurtosis										
1	0.020	0.017	0.025	0.005	0.652	0.009	−0.002	−0.001	−0.010	0.263
2	0.022	0.018	0.027	0.004	0.645	−0.003	−0.003	0.007	0.010	0.187
3	0.027	0.022	−0.016	−0.043	0.004	0.007	−0.003	−0.019	−0.026	0.047
diff	0.008	0.005	−0.041			−0.002	−0.001	−0.018		
<i>p</i> value	0.351	0.526	0.013			0.738	0.938	0.201		
Panel B: DPUV Methodology										
Skewness										
1	0.018	0.020	0.007	−0.011	0.359	0.000	−0.002	−0.018	−0.018	0.093
2	0.026	0.021	0.022	−0.005	0.633	0.001	0.001	0.001	−0.000	0.968
3	0.037	0.014	0.028	−0.009	0.402	0.008	−0.006	0.008	−0.001	0.915
diff	0.019	−0.006	0.021			0.008	−0.004	0.025		
<i>p</i> value	0.023	0.487	0.133			0.238	0.527	0.039		
Kurtosis										
1	0.035	0.021	0.027	−0.008	0.463	0.011	−0.001	0.006	−0.005	0.578
2	0.027	0.016	0.019	−0.008	0.418	0.001	−0.001	−0.004	−0.005	0.543
3	0.017	0.022	0.014	−0.003	0.776	−0.001	−0.003	−0.003	−0.002	0.838
diff	−0.018	0.001	−0.013			−0.012	−0.001	−0.009		
<i>p</i> value	0.023	0.928	0.309			0.083	0.831	0.430		

Note: The table above presents the returns of portfolios which are double sorted on risk-neutral moments. We independently sort firms into terciles based on their implied volatility, then sort these portfolios into terciles based on either implied skewness or implied kurtosis. As within table 3, firms' implied moments are estimated using options with a time-to-maturity nearest to 3-months. We estimate implied moments using two methodologies, in Panel A following the approach of Conrad *et al.* (2013), and in Panel B following the approach of DeMiguel *et al.* (2013). The left side of the table reports mean cumulative quarterly (3-month) returns of the double-sorted portfolios, and the right-side reports mean characteristic adjusted quarterly returns using Fama and French (1993) 5 × 5 size and book-to-market portfolios to benchmark. We also report the difference in returns between the top and bottom skewness or kurtosis tercile portfolios for the given volatility tercile within the 'diff' row. Thus, the 'diff' result represents the quarterly returns (raw or characteristic adjusted) of a long-short, zero net-cost portfolio long high skewness (or kurtosis) firms and short low skewness (or kurtosis) firms within the given volatility tercile portfolio. *P*-values of 'diff' portfolio returns are reported in the row below the 'diff' estimate with the '*p* value' row heading.

simulated, investors earn a risk-premium for holding high implied volatility portfolios (instead of low implied volatility portfolios), resulting in a +50 bp per quarter raw return (+20 bp characteristic adjusted return). However, when using the Conrad *et al.* (2013) methodology the same long-short portfolio sorted on implied volatility actually loses −30 bp a quarter (−50 bp characteristic adjusted return), representing a negative risk-premium for holding higher implied volatility firms— a result which is hard to justify economically. Yet, even in this example highlighted the statistical significance is weak, so one could plausibly argue that the differences may not be methodology dependent. But, this argument is quickly refuted once one examines results of double-sorted portfolios.

When examining the double-sorted portfolios, more significant differences in the implications for the cross-section of stock returns using each approach become apparent. Examining the 12-month maturity in table 6, we show the results of double, independent sorts which first sort on implied volatility and then implied skewness or implied kurtosis. Tests

for differences in the extreme portfolios holding one of the sorts constant are also reported. Both raw and characteristic adjusted returns reported within the table show results which vary considerably using the Conrad *et al.* (2013) and DeMiguel *et al.* (2013) approach. For example, within the low volatility tercile portfolio, there is a negative difference in cumulative quarterly return between the high and low skewness portfolio (−70 bp per quarter) using the Conrad *et al.* (2013) approach and a positive difference (+290 bp a month) in cumulative quarterly return between the same portfolios using the DeMiguel *et al.* (2013) approach. The difference in results under methodologies is about 360 per quarter, indicating a roughly 14.4% annual difference in long-short portfolio returns based on the choice of method to estimate implied moments. Again, this stark difference in results based on methodology, which is also statistically significant at a *p*-value of .05, strongly suggests that the implications for the cross-section of stock returns under the Conrad *et al.* (2013) and DeMiguel *et al.* (2013) approaches vary in a complex manner. Thus, we argue that the results

Table 6. Risk-neutral moment independent double-sorted portfolios – 12 Months.

Vol	Return					Char-Adj Return				
	1	2	3	diff	p value	1	2	3	diff	p value
Panel A: CDG Methodology										
Skewness										
1	0.022	0.011	−0.020	−0.042	0.005	−0.005	−0.003	−0.033	−0.029	0.028
2	0.018	0.027	0.018	−0.000	0.992	0.001	0.003	−0.004	−0.005	0.534
3	0.015	0.040	0.028	0.013	0.254	0.008	0.018	0.002	−0.006	0.494
Diff	−0.007	0.029	0.048			0.013	0.020	0.035		
p value	0.397	0.001	0.004			0.060	0.005	0.014		
Kurtosis										
1	0.022	0.034	0.021	−0.001	0.901	0.005	0.010	0.000	−0.005	0.561
2	0.018	0.028	0.014	−0.004	0.668	0.001	0.007	−0.012	−0.012	0.123
3	0.020	0.017	−0.001	−0.021	0.234	−0.003	−0.000	−0.021	−0.018	0.230
Diff	−0.002	−0.017	−0.022			−0.008	−0.011	−0.021		
p value	0.765	0.054	0.245			0.214	0.147	0.187		
Panel B: DPUV Methodology										
Skewness										
1	0.016	0.009	−0.004	−0.020	0.288	−0.003	−0.011	−0.026	−0.022	0.162
2	0.016	0.025	0.001	−0.016	0.115	−0.001	0.002	−0.021	−0.021	0.013
3	0.045	0.034	0.036	−0.010	0.511	0.007	0.014	0.013	0.006	0.630
Diff	0.029	0.025	0.040			0.010	0.025	0.039		
p value	0.023	0.003	0.049			0.402	0.001	0.024		
Kurtosis										
1	0.035	0.029	0.027	−0.008	0.673	0.010	0.004	0.005	−0.005	0.753
2	0.027	0.023	0.009	−0.018	0.097	0.005	0.003	−0.016	−0.021	0.020
3	0.012	0.021	0.007	−0.006	0.817	−0.006	−0.003	−0.015	−0.009	0.687
Diff	−0.023	−0.008	−0.021			−0.016	−0.007	−0.020		
p value	0.201	0.370	0.437			0.307	0.393	0.389		

Note: The table above presents the returns of portfolios which are double sorted on risk-neutral moments. We independently sort firms into terciles based on their implied volatility, then sort these portfolios into terciles based on either implied skewness or implied kurtosis. As within table 4, firms' implied moments are estimated using options with a time-to-maturity nearest to 12-months. We estimate implied moments using two methodologies, in Panel A following the approach of Conrad *et al.* (2013), and in Panel B following the approach of DeMiguel *et al.* (2013). The left side of the table reports mean cumulative quarterly (3-month) returns of the double-sorted portfolios, and the right-side reports mean characteristic adjusted quarterly returns using Fama and French (1993) 5 × 5 size and book-to-market portfolios to benchmark. We also report the difference in returns between the top and bottom skewness or kurtosis tercile portfolios for the given volatility tercile within the 'diff' row. Thus, the 'diff' result represents the quarterly returns (raw or characteristic adjusted) of a long-short, zero net-cost portfolio long high skewness (or kurtosis) firms and short low skewness (or kurtosis) firms within the given volatility tercile portfolio. *P*-values of 'diff' portfolio returns are reported in the row below the 'diff' estimate with the 'p value' row heading.

of Conrad *et al.* (2013), and other leading papers which utilize the approach of Dennis and Mayhew (2002), should be reconsidered using the much more accurate methodology of DeMiguel *et al.* (2013).

The results of all of our empirical tests suggest that there will be differences in inferences made, in some cases slight and in some quite serious, which are dependent on the choice of methodology to extract BKM implied moment estimates. The biases that are introduced in each approach are due to market frictions and options market microstructure and have nothing to do with firms' implied return distributions. From our broad simulations and limited empirical study, it is clear that the results using the two-step interpolation approach of DeMiguel *et al.* (2013) are much more accurate and robust than those using the approach of Dennis and Mayhew (2002), which is followed by Conrad *et al.* (2013) and others. This, of course, implies that it may be wise to reconsider the results of leading studies including Conrad *et al.* (2013) using the methodology of DeMiguel *et al.* (2013).

5. Conclusion

We examine the two leading approaches to estimate the BKM implied moments taking into account violations in the underlying assumptions due to imperfect markets. The two approaches we examine are the more simplistic estimation approach of Dennis and Mayhew (2002) used by Conrad *et al.* (2013), and the more sophisticated two-step interpolation approach of DeMiguel *et al.* (2013). While each approach has its potential benefits and drawbacks, our simulation experiments definitively prove that the latter approach is much more robust, accurate, and suitable for use in practical applications. The simplistic approach used by Conrad *et al.* (2013) is exposed to significant biases in implied moment estimates such that the method is likely to be an unreliable estimator of the firm's implied density. Even more concerning, new and very popular cross-sectional inferences of firms' returns based on BKM estimates are actually wildly sensitive to firms' varying options microstructures, and as such, are likely

proxies for omitted variables unrelated to a firm's return distribution.

Considering this, we recommend that the approach of DeMiguel *et al.* (2013) is used in future studies by academics and practitioners whose work necessitates accurate estimation of option-implied densities. We also note that the model-free implied volatility estimate of Fukasawa *et al.* (2011) matches the near perfect accuracy of the DeMiguel *et al.* (2013) approach under all simulations tested, and is thus a favorable alternative approach if one is only concerned with estimating implied volatility.

Finally, based on the considerable differences in results we demonstrate under the Dennis and Mayhew (2002) and DeMiguel *et al.* (2013) approaches conducting a limited replication of Conrad *et al.* (2013), it appears wise to reevaluate the empirical results of major studies examining option-implied moments and the cross-section of stock returns using the more robust, consistent, and less-biased approach of DeMiguel *et al.* (2013).

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